Case E

Ms. Lee — Thoracic Outlet Syndrome | Working Dx: Neurogenic/Vascular Mixed TOS

Target: 70%+ | Musculoskeletal (MSK) Physiotherapy

■ PROBLEM LIST

- Right brachial plexus neurodynamic sensitisation
- Pectoralis minor and anterior scalene hypertonicity
- Reduced first rib mobility
- Forward head posture and rounded shoulder girdle
- Intermittent subclavian vascular compromise
- Possible thoracic hyperkyphosis compressing outlet
- Unable to perform overhead reaching at work without symptoms
- Difficulty sustaining repetitive upper limb tasks at end of shift
- Reduced right hand grip for packing/lifting
- Possible activity avoidance due to hand colour changes
- Impact on productivity and job security

■ RED FLAGS

- Persistent Hand Pallor / Cyanosis (Vascular Crisis)
- Progressive Muscle Wasting / Intrinsic Hand Weakness
- Bilateral Upper Limb Symptoms / Cervical Myelopathy Signs
- Unexplained Weight Loss / Apical Lung Mass (Pancoast Tumour)
- Signs of DVT / Axillary-Subclavian Vein Thrombosis (Paget-Schroetter)

■ PATHOPHYSIOLOGY

- 2 Pathophysiology of Working Diagnosis Working Diagnosis: Thoracic Outlet Syndrome — Neurogenic/Vascular Mixed Type The thoracic outlet is a narrow triangular space bounded by the anterior and middle scalene muscles, the first rib, and the clavicle.
- The brachial plexus (C5–T1), subclavian artery, and subclavian vein all pass through this space, making any reduction in its dimensions clinically significant.
- Lee's case, hypertrophy and hypertonicity of the anterior scalene muscle — consistent with her repetitive overhead work — narrows the scalene triangle, compressing the lower trunk of the brachial plexus (C8/T1) and intermittently compressing the subclavian artery.
- This produces the mixed neurogenic-vascular presentation we see clinically.
- Tightness of pectoralis minor, confirmed on examination, creates a secondary compression point in the subcoracoid space.

■ DIFFERENTIAL DIAGNOSES

1. Supporting features:

- Thoracic Outlet Syndrome — Neurogenic/Vascular Mixed Type
- Supporting features:

2. Supporting features:

- Cervical Radiculopathy (C8–T1 Nerve Root Compression)
- Supporting features:
- Age-appropriate (though younger presentation), numbness and paraesthesia in upper limb, possible weakness of intrinsic hand muscles (C8/T1 territory), worsening with upper limb loading.

3. Supporting features:

- Diabetic Peripheral Neuropathy
- Supporting features:
- Differentiating factors:

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Roos Stress Test (EAST: Elevated Arm Stress Test)
What it measures: Neurovascular compromise at the thoracic outlet under sustained provocative positioning; assesses both neurogenic and vascular TOS simultaneously. | How to perform: 1. Patient sits upright. Instruct them to abduct both shoulders to 90°, externally rotate maximal

Adson's Maneuver (Scalene Compression Test)
What it measures: Subclavian artery compression through the scalene triangle; assesses the arterial TOS component specifically. | How to perform: 1. Palpate and monitor the radial pulse on the affected right side. 2. Ask the patient to take a deep breath in and hold it (increases

Upper Limb Neurodynamic Test 1 (ULNT1 / Median Nerve Bias)
What it measures: Mechanosensitivity of the median nerve and brachial plexus; detects neural tension and sensitisation along the neurodynamic chain from cervical spine to hand. | How to perform: 1. Patient supine. Depress the shoulder girdle (scapular depression). 2. Abduct the a

■ TREATMENT PLAN

■ Initial Phase — Weeks 1–4: Decompression, Neural Mobility & Postural Correction

- "Your symptoms are caused by compression of nerves and blood vessels in your shoulder region — this is very manageable with physiotherapy and postural changes." — Provide a clear, non-threatening explanation to prevent catastrophising and empower self-management.

• "We're going to work on your posture and the tight muscles around your shoulder to take pressure off those structures — you can actively contribute to your own recovery." — Linking treatment to mechanism builds adherence and buy-in.

• "Please rest your arm immediately if your hand turns blue or pale and does not recover within a few minutes — this is a safety sign that needs medical attention." — Critical safety education specific to vascular TOS; empowers appropriate help-seeking.

• "At work, try to take regular microbreaks, vary your posture, and avoid sustained overhead positions for more than 2–3 minutes without rest." — Modifiable occupational exposure is the primary driver — load management is central to recovery.

• VAS reduced to ≤3/10 at rest and during usual work tasks

■ Long-Term Plan — Weeks 4–12: Strength, Endurance & Work Rehabilitation

• Serratus anterior and lower trapezius strengthening — upward rotation of the scapula widens the thoracic outlet during overhead activity; these muscles are essential for dynamic stability and shoulder girdle positioning during Ms. Lee's warehouse tasks.

• Thoracic mobility and postural endurance training — address thoracic kyphosis contributing to chronic outlet compromise; progress to sustained posture challenge in work-simulated positions.

• Graduated return-to-work planning — collaborate with Ms. Lee and employer on workstation modification (height-adjustable surfaces, task rotation), progressive exposure to overhead tasks with monitoring of symptoms using the DASH and Roos re-test.

• Cardiovascular fitness and weight management — low-impact aerobic exercise to improve peripheral circulation and reduce adipose tissue load on the outlet. Coordinate with her endocrinologist to ensure exercise safety with Type 1 Diabetes (glucose monitoring, hypoglycaemia awareness).

• Ongoing glycaemic management education — encourage tight blood glucose control as this directly affects peripheral nerve health and vascular compliance. Refer to diabetes nurse/dietitian as needed.